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Eid in Makkah

Course: Computer Graphics CS2206

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Introduction

1.1.description of the general idea

This project aims to design and implement a C++ graphical application using OpenGL that captures the essence of Eid al-Fitr in Mecca through an interactive visual experience. The core idea is to recreate the festive environment by modeling scenes that reflect the joy, unity of Eid Al-Fitr. The application contains key computer graphics principles such as object rendering, scene transformation, and basic animation presenting a creative representation of a meaningful cultural celebration using modern graphics techniques.

1.2.goal

The goal of this project is to write a complete C++ application using OpenGL to visually showcase Eid al-Fitr in Makkah. The project aims to simulate key elements of the celebration, Salat al-Eid and the happines feeling in people through interactive 2D/3D graphics.

Also to use core computer graphics concepts such as rendering, transformations (translation, scaling, and rotation) and user interaction through keyboard and mouse buttons. The project highlights how programming and graphics can be combined to represent real-world cultural (Eid Al-Fitr) experiences in a digital environment (OpenGL).

1.3.contributions of the project

In this project we showcased the first morning of Eid al-Fitr in Makkah with the family members (father : Abdul-allah, mother : Sarah, and daughter (Main character): Rafef) living in Makkah. The project contains moments starting from Rafeef waking up with her family and getting ready for salat al-Eid to Rafef and her mom going to al-Mosalla and the mother answering Rafef questions, to the family gathering outside the Mosque and Rafef holding a basket with dates in it to finally Rafef jumping from happiness after understanding how Eid in Makkah is so lovely.

Methodology

2.1. Implementation Details and Functionaliy

The core technical objective of this project is to simulate the cultural and spiritual atmosphere of Eid al-Fitr in Makkah through an interactive graphical environment. This was achieved by utilizing the C++ programming language in conjunction with the OpenGL (Open Graphics Library).

Core Functionalities: The following technical functionalities were implemented to ensure a smooth user experience:

- Scene Rendering: Using OpenGL's rendering pipeline, we constructed four distinct scenes depicting the morning of Eid.

- Geometric Transformations: To animate the characters and environment, fundamental transformations were applied:

- Translation: Used for moving the mother and daughter characters across the screen and for the jumping animation.
- Rotation: Applied to animate objects such as the sun’s rays, while the clock hand movement was implemented using mathematical calculations based on angle updates.
- Scaling: Used to adjust the sizes of characters and objects to create a sense of depth and perspective.
- Texture Mapping: The project utilizes 2D textures to enhance realism, specifically for the wooden floor, the wardrobe, the prayer mat (sujadd), and the traditional shomaq.
- Interactive Controls:
- Mouse Input: Used to navigate between the four scenes.

- Keyboard Input: Allows the user to move the mother character (using 'A' and 'D' keys) and trigger the daughter's jumping animation (using the 'M' key).
- Animation and Timing: A timer function was implemented to manage real-time updates for the jumping physics (gravity simulation) and the continuous rotation of the sun and clock.

Additional Libraries Used:

- GLUT/FreeGLUT: Utilized for managing the window system, handling input events (mouse and keyboard), and managing the main loop and timer functions.

By focusing on these core OpenGL principles, the project successfully creates a cohesive digital story that combines technical computer graphics with cultural narrative.

Result

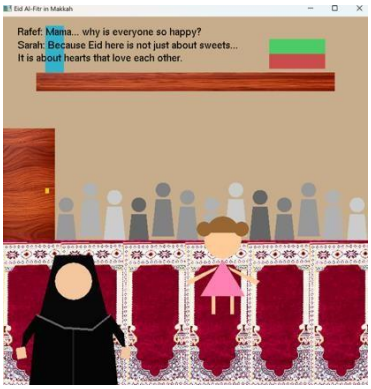
In the first scene we added a 3d object and texture for the wardrobe,the vanity, we also animated the clock, and We added a color gradient to the mirror.

Forfuture Extend we can add some motion for the character to make them talk and walk.



In the second scene we made it looking like a mosque, and added a crowd in the background, also we animated mother to move toward to the main character,also we added texture to the door and shelf.

For future Extend we might move the main characters toward the mother and make some conversation between the crowd and the main characters.



In the third scene the main character and the mother go outside the mosque and they distribute dates to children and passersby while they wait for the father to come out.
For future Extend we can add the mosque in the background and include side characters that take Dates from the main character.



In the fourth scene, after father arrived, they have a conversation while walking towards the hotel. the main character starts jumping when she realize the meaning of Al Eid in Makkah.

For future Extend we might add a car and make it move toward the hotel.



Conclusion & future work

In this project, we designed and built 2D graphical scenes using OpenGL. The work includes both indoor and outdoor environments with different objects and characters. We also added a simple 3D element to make the scene look more realistic.

Through this project, we were able to understand how separate components such as shapes, textures, and transformations work together to form a visual experience.

This project helped us to better our understanding of how to combine different graphical elements to create a complete scene. It also improved our ability to represent real-world ideas using graphical techniques.

For future work, we can improve the scenes by adding, for instance, more realistic details such as lighting effects, shadows, and better-looking textures. We can also add more animated moves to improve the movement of the characters to make it more smooth and more natural. Another possible improvement is adding more scenes. Finally, adding more 3D objects can make the scene more detailed.

Reference

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